

CASM: CONTEXT-AWARE SEMI-MARKOV POST-PROCESSOR FOR BEAT TRACKING

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ABSTRACT

At the final stage of beat tracking, neural activations need to be decoded into a discrete, musically coherent event sequence. Direct peak picking follows the local evidence but tends to produce spurious and missed detections. Dynamic Bayesian networks (DBNs), the dominant post-processing approach, suppress such errors by imposing explicit human-specified global assumptions. They work well when predefined tempo and meter assumptions match the music, but expressive or rhythmically complex pieces often depart from them, resulting in overrides correct local evidence instead of correcting it. We address this limitation by proposing CASM, a model-agnostic, context-aware semi-Markov decoder that exploits local structure in beat activations. CASM estimates local periodicity and ambiguity from the activations, then commits strongly when the cues are clear and relaxes toward the neural output when they are ambiguous, with built-in safeguards keeping this behavior stable and predictable. Using a single configuration, CASM improves F1, CMLt, and AMLt across BeatThis, MSCNN, and TCN models on GTZAN, and delivers state-of-the-art performance on SMC. Further analysis shows that CASM is less sensitive than DBNs to the choice of calibration data.¹

Index Terms— beat tracking, context-aware, semi-Markov model, post-processing, local periodicity

1. INTRODUCTION

Beats provide a temporal scaffold for following and organizing music. Successful beat tracking is crucial for music production, structural analysis, synchronization, and interactive systems [1, 2]. Despite recent attempts to remove post-processing from neural beat tracking [2, 3], many modern systems still rely on structured decoders such as dynamic Bayesian networks (DBNs) to obtain musically coherent beat estimates [4]. Even with increasingly accurate neural activations, framewise predictions do not directly guarantee a coherent event sequence: direct peak picking can produce duplicated or missing beats, metrical-level switches, or downbeats that disagree with the beat grid. Post-processing therefore remains useful for converting local activation evidence into a temporally consistent sequence. The central difficulty is how strongly this temporal structure should be imposed: weak decoding follows the network closely but can sacrifice continuity, whereas strong decoding can override correct evidence under expressive or unusual timing.

DBNs decode a path through tempo, phase, and meter states under assumptions specified before the track is observed [4, 5]. This

is effective when assumptions match the input, but corpus-specific retuning can become necessary for music outside these global assumptions [6]. PLPDP provides a more local alternative: it estimates a time-varying inter-beat interval and confidence from predominant local pulse (PLP), and uses them to adapt the temporal constraint during dynamic programming [1]. This establishes an important principle for activation decoding: the temporal prior can depend on the local evidence itself. However, the strongest local period need not be decisive. Half- and double-tempo hypotheses can receive similar support, in which case strongly enforcing either interpretation may be undesirable.

We introduce CASM, a context-aware semi-Markov decoder that makes this distinction explicit. CASM constructs a sparse path over activation-supported peaks and estimates both local periodicity and its ambiguity. These estimates control the target, strength, and tolerance of the segment-duration preference: when the periodic evidence is clear, CASM commits strongly to local regularity; when competing periods are similarly plausible, it weakens that preference and lets the neural activations dominate. Fig. 1 illustrates both cases. On the left, clear local periodicity allows CASM to recover weak, activation-supported beats that Direct misses; on the right, competing periodicity hypotheses make CASM defer to the Direct path. A beat-count safeguard and beat-synchronous meter decoder further keep the procedure deterministic and predictable.

CASM is a model-agnostic plug-in needing no backbone retraining or per-track BPM/meter tuning, and its single calibrated configuration is less sensitive than DBNs to the choice of calibration data.

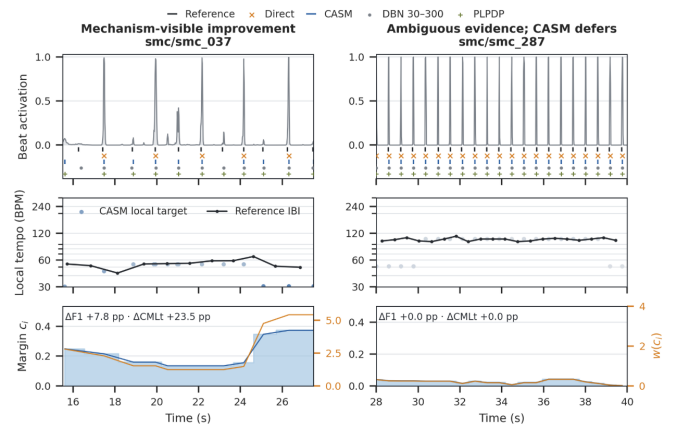


Fig. 1. CASM behaviour on held-out Beat This activations from SMC. Clear local periodicity recovers weak activation-supported beats (left), whereas ambiguous competing periods make CASM defer to Direct (right). Reference IBI is shown only for interpretation.

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¹Code and demo are available at <https://zhanh-he.github.io/casm-beat-tracking>.

2. RELATED WORK

2.1. Structured post-processing

Dynamic programming has long balanced onset evidence against inter-beat regularity [7]. Bayesian beat-position models used a dominant beat interval [8], while learned CRFs were later extended from beats to downbeats, long-range section consistency, and genre-specific microtiming [9–12]. Resonating comb filters estimate a dominant periodicity before event alignment [13], and DBNs enumerate tempo–phase–meter states and decode their most likely trajectory [4, 5]. These methods differ in state representation and optimization, but they share a useful principle: noisy framewise evidence should be interpreted as a sequence rather than thresholded independently.

2.2. From PLPDP to ambiguity-aware local constraints

PLPDP is CASM’s direct methodological predecessor [1]. Its central diagnosis was that a fixed global tempo penalty, including the empirically chosen transition law of common HMM trackers, can be too rigid for expressive timing. PLPDP extracts a time-varying inter-beat interval and a confidence curve from PLP, then replaces the global target and weight of dense-frame dynamic programming with those local quantities. Its real-time continuation improves controllability [14], and dPLP makes local-pulse estimation differentiable [15]. Reliability-informed tracking previously established the broader principle that temporal commitment should depend on evidence quality [16]. CASM’s margin is a local, deterministic separation score rather than predicted track-level accuracy. It also differs from the causal one-dimensional semi-Markov state space of [17], which maintains a framewise posterior and adapts a jump-back distribution online. CASM is an offline maximum-score path over candidate-to-candidate segments. Nor is it a semi-CRF: its potentials are hand-specified, unnormalized, and not learned [18].

2.3. Alternatives beyond plug-in decoding

Particle filters maintain online tempo hypotheses [19], and BeatNet/BeatNet+ extend this branch to joint real-time rhythm analysis [20, 21]. Another line adapts a tracker with user annotations [22–26]. More recent systems remove or redesign post-processing: masked diffusion iteratively refines coherent sequences [27], BeatFCOS predicts intervals as objects with Soft-NMS [28], and other work reformulates the prediction target [29]. These approaches change the predictor or import additional supervision. CASM instead addresses the deployment setting in which only the activation output of a frozen tracker is available.

3. METHODOLOGY

3.1. CASM Overview

The proposed Context-Aware Semi-Markov (CASM) decoder is a model-agnostic post-processor that operates on beat and downbeat activations produced by a frozen tracker. It modifies only the decoding stage and therefore requires no retraining. Like PLPDP, CASM estimates a local beat period and its support from the activation sequence. CASM differs in how this support is defined and used: it measures the separation between the dominant period and competing tempo hypotheses and uses this margin to adapt the rhythmic constraint. Consequently, CASM enforces local regularity when the

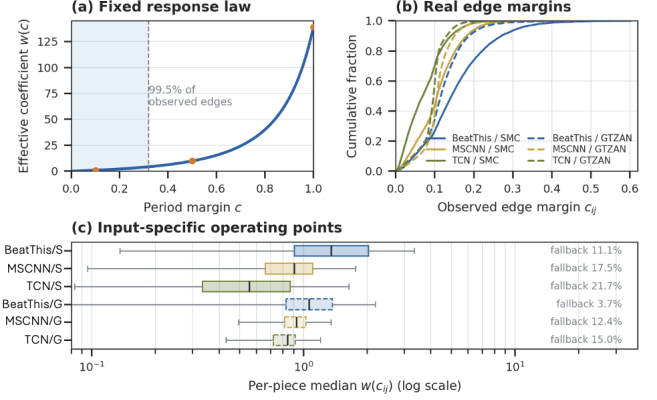


Fig. 2. Input-conditioned duration stiffness under one frozen CASM configuration. (a) The fixed constants map period margin c to coefficient $w(c)$; shading contains 99.5% of observed structured-path edges. (b) Edge margin distributions differ across backbones and corpora. (c) Per-piece median coefficients therefore occupy different operating ranges; annotations give the separate count-ratio fallback rate. Fixed parameters define one response law, not one effective rigidity for every input.

periodic evidence is clear, but relaxes the constraint when the evidence is ambiguous.

3.2. Context-aware beat decoding

A frozen tracker produces a beat logit z_t^b at each frame t , with activation $a_t = \text{sigmoid}(z_t^b)$ and frame rate F . We retain local maxima above a fixed threshold θ_c as candidates $\mathcal{C} = \{t_1, \dots, t_N\}$, assigning each candidate a clipped evidence score e_i . For $i < j$, the elapsed time is $\Delta_{ij} = (t_j - t_i)/F$, and edge (i, j) is admissible when $60/\Delta_{ij}$ lies within 30–300 BPM.

CASM selects an ordered candidate path $\pi = (i_1, \dots, i_K)$, with $i_1 < \dots < i_K$, by maximizing

$$\mathcal{S}(\pi) = \sum_{k=1}^K e_{i_k} - \sum_{k=2}^K D_{i_{k-1}, i_k} \quad (1)$$

where D_{ij} is the duration cost between candidates i and j . The two terms respectively reward strong beat evidence and penalize rhythmically inconsistent intervals. Because each transition scores a variable-duration interval, CASM has a semi-Markov structure.

At candidate i , $q_i(\ell)$ denotes the normalized autocorrelation score at an admissible lag ℓ within a window centred on t_i , including a mild fast-tempo prior. The dominant lag $\ell_i^* = \arg \max_{\ell} q_i(\ell)$ gives the local period $\tau_i = \ell_i^*/F$. We define the strongest competing hypothesis outside a ± 2 -frame neighbourhood as $q_i^{\text{alt}} = \max_{\ell: |\ell - \ell_i^*| > 2} q_i(\ell)$. The normalized period-separation margin is

$$c_i = \text{clip}_{[0,1]} \left(\frac{q_i(\ell_i^*) - q_i^{\text{alt}}}{|q_i(\ell_i^*)| + \epsilon_c} \right) \quad (2)$$

where $\epsilon_c > 0$ stabilizes the denominator. A large c_i indicates a clearly dominant period, whereas a small value indicates ambiguity, including half- or double-time alternatives. Thus, c_i is a deterministic separation margin rather than a calibrated probability. Both τ_i and c_i are median-filtered over five consecutive candidates.

For edge (i, j) , CASM combines the endpoint estimates as $\bar{\tau}_{ij} = \sqrt{\tau_i \tau_j}$ and $c_{ij} = \sqrt{c_i c_j}$. The ambiguity-conditioned con-

straint strength is $g(c) = \lambda c / \{2[\sigma_0 + (1 - c)\sigma_u]^2\}$, giving the duration cost

$$D_{ij} = g(c_{ij}) \log^2 \left(\frac{\Delta_{ij}}{\bar{\tau}_{ij}} \right) \quad (3)$$

Here, $\lambda > 0$ controls the overall penalty strength, $\sigma_0 > 0$ sets the minimum timing tolerance, and $\sigma_u \geq 0$ controls relaxation under ambiguity. The cost vanishes when $\Delta_{ij} = \bar{\tau}_{ij}$; a large c_{ij} enforces rhythmic regularity, while a small c_{ij} relaxes the constraint. CASM can therefore recover weak but rhythmically supported beats without forcing an unreliable tempo grid.

Figure 2 shows how Eq. (3) converts the local period-separation margin into an input-dependent timing constraint. Panel (a) shows the fixed response, panel (b) maps transitions from different datasets and backbones onto it, and panel (c) summarizes the resulting track-level constraint strengths. Thus, one frozen CASM configuration adapts its constraint across inputs. The figure illustrates decoder conditioning, not tracking accuracy; the direct fallback is a separate safeguard.

3.3. Safeguards and downbeat decoding

A Viterbi-style dynamic program maximizes Eq. (1) and backtracks the best beat path, allowing a restart after an unfavourable prefix. As a safeguard, CASM also forms a direct fallback path using seven-frame max pooling, a zero-logit threshold, and one-frame deduplication [2]. If $|\pi_{\text{CASM}}|/|\pi_{\text{direct}}| \notin [r_{\min}, r_{\max}]$, indicating an implausibly sparse or dense CASM path, the direct path is returned. This rule is deterministic and requires no learned router.

Downbeats are decoded on the selected beat grid using a second dynamic program that favours valid bar lengths and discourages unnecessary meter changes. If its output strongly disagrees with the direct downbeat activations snapped to the same grid, the snapped result is used, ensuring that every downbeat coincides with an output beat. All scalar parameters are calibrated once on the development folds and then frozen across tracks, datasets, and backbone models. Inference requires no reference annotations, externally supplied BPM or meter, per-track tuning, or learned decoder parameters.

4. EXPERIMENTS

4.1. Evaluation protocol

Datasets and splits. We use the publicly released data from BeatThis [2], which contains 4,556 tracks from 18 beat-tracking datasets, including SMC, Hainsworth, Ballroom, Beatles, and others. This data collection provides preprocessed log-Mel spectrograms computed from 22.05kHz mono audio at 50 fps, together with a fixed eight-fold partition to conduct cross-validation experiments. For each fold, the corresponding backbone checkpoint is trained on the remaining seven folds, ensuring that evaluated tracks remain unseen during training. Among these datasets, SMC provides beat but not downbeat annotations, and is widely recognized as the hard one compared to the others. Additionally, GTZAN provides 993 tracks as the held-out evaluation set, which is excluded from the cross-validation pool.

Backbones. We select three models whose implementations are publicly available and readily reproducible. For BeatThis, we use the original code and pretrained checkpoints [2]. MSCNN is adapted from the multi-scale, multi-task convolutional network of [30], which was designed for 60s Bark-scale specific-loudness features. We modify it to accept the 30s the unified log-Mel input and remove the task heads unrelated to beat and downbeat tracking.

TCN was introduced by [31], and we use the re-implementation distributed with [30], which has been adapted from its original 5 s input to the same 30 s log-Mel input, with the tempo task head removed. All resulting backbones are trained on the same data and frozen before decoding performance comparison, ensuring that every post-processor receives exactly the same 50 Hz framewise beat/downbeat activations from a given backbone.

Evaluation. Following previous work [2], the eight fold-specific checkpoints of each backbone are evaluated on GTZAN and their corresponding out-of-fold SMC partitions, with scores macro-averaged over total evaluation. We use `mir_eval` to compute F1 score with a ± 70 ms tolerance, CMLt, and AMLt [32]. CMLt measures continuity at the correct metrical level, whereas AMLt additionally accepts standard off-beat and half- or double-level interpretations.

4.2. Post-processors and tempo support

As we defined our task is offline, the future information is accessible for model to top up the estimate accuracy. We compare CASM with the direct decoding, along with other offline post-processing methods including DBN [4], PLPDP [1], CombFilter [13], and CRF beat detection [8]. Direct is the peak-picking path defined in Sec. 3, and CASM uses its fixed 30–300-BPM support. PLPDP, CombFilter, and CRF are run with their released/default parameters. For methods without a joint beat/downbeat decoder, downbeats are decoded from the downbeat activation stream and snapped to the corresponding beat grid.

DBN receives separate treatment because its behaviour is particularly sensitive to global tempo support. We use the standard joint madmom configuration [4]: meters $\{3, 4\}$, transition weight 100, observation weight 16, threshold 0.05, and the default 55–215-BPM range. Since many SMC data are fall below 55 BPM [6], Table 1 also explores a 30–300-BPM DBN variant in which only the tempo range is changed. For a matched diagnostic, we additionally report CASM with its support restricted to 55–215 BPM on BeatThis activations, with all other CASM settings remain unchanged.

4.3. CASM calibration

CASM has no learned weights or random seed. The final configuration used in all main tables is calibrated once on the union of SMC folds 1–7 (7F) and then frozen for every track, backbone, and dataset. The same global scalar settings are therefore used on BeatThis, MSCNN, and TCN, on both SMC and GTZAN, without per-track or per-dataset BPM, meter, or parameter tuning at inference. Because the aggregate SMC results use eight-fold out-of-fold backbone activations while the decoder configuration is calibrated on folds 1–7, they are out-of-fold with respect to backbone training but are not a nested estimate of decoder calibration.

We analyse calibration-data sensitivity separately rather than using it to choose the main operating point. SMC fold 0 is held out from decoder calibration throughout. From folds 1–7, we enumerate every 1-, 2-, and 4-fold calibration subset (7, 21, and 35 combinations) and the seven-fold union. For both CASM and DBN, a decoder configuration is selected independently on each calibration subset and then evaluated on the same fixed SMC-fold-0 panel and on GTZAN using one fixed backbone seed. The resulting distributions therefore measure sensitivity to the amount and composition of calibration data, while the 7F CASM configuration is the maximally supported global configuration used for the main results.

Table 1. Comparison with post-processing methods on fixed backbone activations. DBN default uses 55–215 BPM, while wide-support variants use 30–300 BPM to cover low-tempo SMC tracks [6], all other settings are unchanged.

Method	BPM	GTZAN (Beat / Downbeat)						SMC (Beat)		
		F1	CMLt	AMLt	F1	CMLt	AMLt	F1	CMLt	AMLt
BeatThis (direct) [2]	–	89.1	79.8	89.8	78.3	67.2	79.1	62.7	51.4	61.0
w. CASM	30–300	+0.3	+0.8	+1.2	+0.7	+3.6	+8.0	+0.3	+2.4	+3.6
w. CASM, ablation	55–215	+0.1	+0.5	+0.6	+0.5	+2.9	+5.8	+0.1	+0.4	+2.0
w. DBN [4]	30–300	–0.5	–0.9	+0.2	–1.5	–0.6	+1.1	–1.6	+0.4	+3.5
w. DBN, default	55–215	–1.0	+0.7	+1.3	–0.9	+6.2	+8.7	–5.2	–3.9	+1.5
w. PLPDP [1]	30–300	+0.1	+0.3	+0.3	+0.1	+0.2	+0.2	–0.9	–3.5	–0.3
w. CombFilter [13]	30–300	–4.0	–3.6	+0.1	–3.3	+3.2	+7.6	–7.1	+1.1	+3.1
w. CRF [8]	30–300	–0.6	–0.3	+1.4	–1.4	+1.9	+5.5	–1.2	+1.4	+1.2
MSCNN (direct) [30]	–	86.1	73.4	79.9	70.8	53.1	63.1	57.2	42.0	48.4
w. CASM	30–300	+0.4	+1.6	+2.4	+1.5	+12.5	+18.3	+0.3	+3.5	+6.8
w. DBN	30–300	–1.1	+0.2	+5.9	–0.9	+9.5	+14.6	–0.2	+2.2	+4.1
w. DBN, default	55–215	–0.6	+4.0	+7.2	+1.8	+15.8	+21.2	–4.9	+0.3	+3.9
TCN (direct) [31]	–	87.1	73.8	81.7	62.4	24.6	59.9	52.9	30.1	36.0
w. CASM	30–300	+0.3	+1.3	+1.6	+2.1	+9.2	+13.4	+1.7	+4.5	+8.2
w. DBN	30–300	–1.9	–0.8	+4.4	–0.2	+8.9	+9.3	–0.4	+2.1	+5.9
w. DBN, default	55–215	–2.6	–0.7	+5.0	+4.9	+23.2	+22.9	–0.9	–7.0	+2.3

Table 2. Comparison with recent beat-tracking systems. BeatThis (direct) and CASM results were obtained from our experiments, while all other values are reported from their original papers.

Method	GTZAN (Beat / Downbeat)						SMC (Beat)		
	F1	CMLt	AMLt	F1	CMLt	AMLt	F1	CMLt	AMLt
HN-MERT w. DBN [33]	89.7	81.4	94.3	77.4	71.6	87.2	61.7	–	–
HN-MusicFM w. DBN	89.2	80.9	93.7	79.8	73.2	89.5	60.6	–	–
BeatFM-MERT w. DBN [34]	89.5	82.7	93.9	76.7	70.2	85.4	61.3	–	–
BeatFM-MusicFM w. DBN	89.1	80.6	93.5	79.6	74.4	88.7	60.5	–	–
BeatMamba w. DBN [35]	88.9	82.3	94.3	–	–	–	58.7	46.8	62.4
BeatKAN w. DBN [36]	88.2	78.1	92.3	–	–	–	59.8	48.4	64.0
MDM-BeatThis (direct) [27]	89.7	82.9	92.5	79.5	76.4	88.5	61.0	48.9	64.3
BeatThis (direct) [2]	89.1	79.8	89.8	78.3	67.2	79.1	62.7	51.4	61.0
BeatThis w. CASM	89.4	80.6	91.0	79.0	70.8	87.1	63.0	53.8	64.6

5. RESULTS

5.1. Comparison with post-processors

Table 1 isolates the post-processor by holding the backbone activations fixed. With 30–300-BPM support, the CASM improves BeatThis (direct pick-peaking) on every reported F1, CMLt, and AMLt entry across all three backbones and both datasets. Relative to DBN, CASM generally preserves F1 better, while DBN can produce larger continuity gains, particularly for GTZAN downbeats. The large SMC change between the 55–215- and 30–300-BPM DBN variants also shows how strongly a fixed DBN can depend on its global tempo range. PLPDP, CombFilter, and CRF provide useful alternatives, but none improves every reported metric under the same fixed activations. Overall, CASM gives a conservative F1–continuity operating point that transfers across backbone families without changing decoder parameters.

5.2. Comparison with the state of the art

Table 2 compares the BeatThis works with CASM results with recent literature. While using CASM improves six beat/downbeat metrics compared to BeatThis (directly peak-picking) on the GTZAN, we reported the highest scores on SMC compared to the other SOTA systems. On GTZAN, stronger task-specific or pre-trained fron-

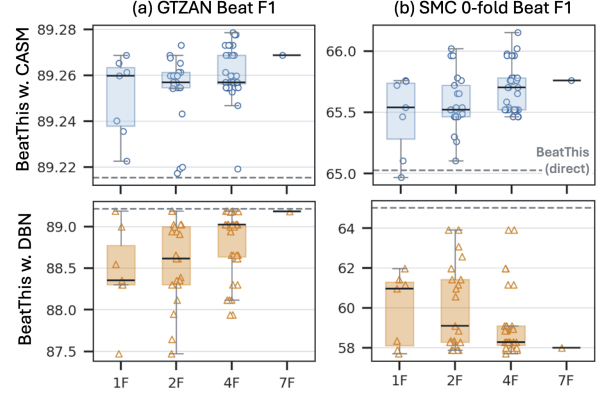


Fig. 3. Sensitivity to calibration data for CASM and DBN. Each point corresponds to a decoder configuration selected from one of the 7/21/35 exhaustive 1-/2-/4-fold subsets of SMC folds 1–7 and evaluated on fixed SMC-fold-0 and GTZAN panels; 7F is a single configuration. Boxes are descriptive because calibration subsets overlap, and dashed lines show Direct performance.

tends retain the best individual scores. The literature table therefore provides system-level context; controlled claims about decoder behaviour come from Table 1, where the backbone activations are held fixed.

5.3. Sensitivity to calibration data

The sensitivity experiment keeps the evaluation panels fixed and changes only the data used to calibrate the decoder. For each calibration scale $k \in \{1, 2, 4, 7\}$, CASM and DBN are calibrated on the corresponding SMC fold subsets described in Sec. 4, and every selected configuration is evaluated on the same held-out SMC fold 0 and the same GTZAN backbone seed. Thus, the spread at a given scale measures dependence on which calibration folds were used, while the movement of the distribution centre measures how additional calibration data changes transfer to the two fixed evaluation panels.

Fig. 3 shows a clear difference in calibration behaviour. CASM remains tightly clustered across fold combinations, and its central performance improves consistently as more calibration folds are used on both the held-out SMC panel and GTZAN. In contrast, DBN exhibits much wider variation across calibration subsets, and the trends on SMC and GTZAN do not move together: configurations favoured by one panel can transfer poorly to the other. This cross-dataset disagreement is consistent with stronger calibration-set over-specialisation in the fixed DBN parameterisation. The 7F CASM setting used in the main tables is therefore supported by the full available calibration pool rather than selected from a favourable smaller-fold combination.

6. CONCLUSION AND FUTURE WORK

CASM revisits structured beat decoding as an inductive bias conditioned on each input, not as a return to a fixed global grid. It joins activation-supported events with locally centred segment potentials and deliberately relaxes those potentials when periodic evidence is ambiguous. Once globally calibrated, the same deterministic module runs without upstream retraining, user annotation, or per-track

tempo and meter settings. Current evidence supports a conservative F1–continuity alternative to Direct and DBN decoding, not universal dominance. The next version should use strict nested or leave-one-dataset-out calibration, matched tempo-support baselines, PLPDP and mechanism ablations, and explicit safeguard statistics. Multi-hypothesis period paths, causal decoding, and a differentiable duration layer are natural extensions, as is evaluation on every recent tracker that releases compatible activations.

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